Review of Enterprise Resource Planning (ERP) Systems

Enterprise resource planning (ERP) is a business software that collects information from different departments to enable leaders to monitor the company with detailed analytics and reporting. It unifies crucial functions such as finance, manufacturing, inventory and order management, sales and marketing, project management, and human resources. It is

In choosing an ERP, businesses must understand the capabilities, implementation models, integration requirements, and total cost of ownership of available software providers before utilizing one.

SAP Basics (SAP SE, 2022)

SAP (System Applications and Products) SE is the market leader for enterprise application software. It is a German multinational company that pioneered software. SAP lets companies streamline processes, use live data, predict customer trends, and connect entire businesses.

SAP was founded in 1972 for the development of real-time data application software. They introduced various versions since then, such as SAP R/2, SAP R/3, SAP ERP, SAP Business Suite 7, SAP HANA, and SAP S/4HANA.

SAP ERP is the standard ERP solution for companies worldwide. It enables a company to support and optimize its business processes and allows the collection of logically related transactions within identifiable business functions.

SAP S/4HANA
SAP's strategic goal has been to develop a next-generation ERP package, such as an integrated management system based on proven business practices and meet the requirements of the digital transformation era consisting of all-encompassing digitization of the economy.

This results in **SAP S/4HANA.** It is the next-generation Business Suite and the biggest innovation since SAP R/3. It is the digital core of the company that enables digital transformation.

SAP ERP is designed to strengthen companies of every size in various industries, while SAP S/4HANA caters more to large, enterprise-level organizations.

SAP Fiori
It is a design system and a user experience (UX) layer that allows business app creation with a consumer-grade user experience. It makes casual users into SAP experts with simple screens that run on any device. SAP Fiori provides a user interface for SAP S/4HANA. It has three (3) application types:

- **Transactional apps** that are used to access tasks like *Create*, *Change*, or *Display Process* with guided navigation;
- **Analytical apps** that give a visual overview of business data and
- A **factsheet** with a view of the essential information about objects and contextual navigation between related objects.

Data Types
These are data types that can be used and found in SAP S/4HANA.

- **Organizational Data** – part of the organizational unit, such as the company code, plant, storage location, and distribution channel.
- **Master Data** – represents logically grouped data like customer master, material master, vendor master, and general ledger accounts.

- **Transaction Data** – the system record of the business event wherein, depending on the business event, different master data and organization data will be referenced.

  The business transaction record is a '**document**,' which includes predefined information from the master data and organization entities. Examples of documents are sales documents, purchasing documents, material documents, and account documents.

Global Bike (SAP SE, 2022)

SAP S/4HANA uses the Global Bike case study in performing their various modules in SAP Fiori. The summary of the story goes as follows:

Global Bike Incorporated has a practical design philosophy from its deep roots in off-road trail racing and long-distance road racing sports. It was founded by John Davis (Frankenstein Bikes) and Peter Schwarz (Heidelberg Composites), who designed their first bikes out of the necessity to win their races as available bikes could not perform to their extremely high standards.

John and Peter share as CEOs, responsible for managing the company's global organization. John is responsible for sales, marketing, service and support, IT, finance, and human resources groups. Peter is responsible for research, design, procurement, and manufacturing groups from an organizational reporting perspective.

*As Global Bike is a process-centric organization, the two prefer to think of the processes they are responsible for rather than the company's functional areas that report to them. Given this, Peter is responsible for **Idea-to-Market** and **Build-to-Stock,** and John is responsible for **Order-to-Cash** and **Service & Support**, as well as the supporting services for all four (4) key processes.*

The following are the observed strategies found in the Global Bike case study:

1. **Product Strategy**: One of the company's product strategies includes having an extensive innovation network to source ideas from riders, dealers, and professionals to improve its bicycles' performance, reliability, and quality.
2. **Manufacturing Strategy**: Global Bike outsources the production of off-road and touring frames and carbon composite wheels to trusted partners with specialty facilities to create the complex materials used. They maintain collaborative research and design relationships with these facilities to ensure that innovations in both material and structural capabilities are included in the frames.
3. **Distribution and Partner Network**: Global Bike sells its bikes exclusively through reputable and respected Independent Bicycle Dealers (IBDs). These dealers have staff members with expertise in off-road and tour racing to help consumers pick the right Global Bike bike and accessories for their individual needs.

   Global Bike establishes an extensive **partner operation** to ensure process continuity between its partners to provide best-in-class products.
4. **IT Strategy**: All ERP functions are centralized to reduce costs and deliver global best-in-class technology to all divisions. This approach gives Global Bike an advanced business platform under a highly controlled environment, enabling consistency of operations and process integrity across the globe.

SAP Modules (SAP SE, 2022)

Review SAP S/4HANA modules and their organizational units, master data, and processes.

Sales and Distribution (SD)
It allows organizations to store and manage customer- and product-related data. The functionalities include sales support, sales, shipping and transportation, billing, credit management, and foreign trade.

Organizational Units:
- **Client** – an independent environment in the system.
- **Company Code** – the smallest organizational unit where a legal set of books can be maintained.
- **Credit Control Area** – grants and monitors a credit limit for customers.
- **Sales Organization** – responsible for the sale of specific products or services. It is also responsible for legal liability for products and customer claims.
- **Plant** – or Delivering Plant, a plant from which the goods should be delivered to the customer.

Master Data:
- **Customer Master** – contains all the necessary information for processing orders, deliveries, invoices, and customer payments.
- **Material Master** – contains all the information a company needs to manage a material.
- **Condition Master** – includes prices, surcharges (additional charge), discounts, freights (goods transferred from place to place), and taxes.

Processes (**Sales Order**):
1. Pre-sales Activities
2. Sales Order Entry
3. Check Availability
4. Pick Materials
5. Pack Materials
6. Post Goods Issue
7. Invoice Customer
8. Receipt of Customer Payment

Materials Management (MM)
It is concerned with helping organizations with material and inventory management and warehouse management in the supply chain process. It also deals with all tasks in the supply chain, such as consumption-based planning, vendor evaluation, and invoice verification.

Organizational Units:
- **Purchasing Organization** – responsible for services and material procurement and in charge of negotiating the conditions of the purchase with the vendors.
- **Purchasing Group** – a key unit representing the buyer or group of buyers responsible for certain purchasing activities. This unit also serves as the communication channel for vendors.

Master Data:
- **Vendor Master** – contains all the necessary information for business activities with an external supplier. It is used and maintained by the Purchasing and Accounting Departments.
- **Purchasing Information Record** – allows buyers to quickly determine which vendors have offered or supplied specific materials.

Processes (**Procure-to-Pay**):
1. Purchase Requisition
2. Vendor Selection
3. Purchase Order
4. Notify Vendor
5. Vendor Shipment
6. Goods Receipt
7. Invoice Receipt
8. Payment to Vendor

Production Planning and Execution (PP)
Production Planning aligns demand with manufacturing capacity in creating production and procurement schedules for the finished products and component materials. As a module, SAP PP tracks and records the manufacturing process flow such as the planned and actual costs and good movement from the conversion of raw material to semi-finished goods.

Organizational Unit:

- **Work Center Locations** – define where and when the operation is performed. These can be machines, people, production lines, or groups of tradespeople.

Master Data:

- **Bill of Materials (BOM)** – lists the components of a product or assembly.
- **Routing** – the series of sequential steps/operations that must be carried out to produce a product.
- **Work Center** – the location within a plant where value-added work (operations or activities) is performed.
- **Product Group** – the aggregate planning that groups materials or other product groups (Product Families).

Processes (**Production Planning**):

1. Forecasting
2. Sales and Operations Planning (SOP)
3. Demand Management
4. Master Production Scheduling (MPS)
5. Material Requirement Planning (MRP)

Processes (**Manufacturing Execution**):

1. Production Proposal
2. Capacity Planning
3. Schedule Planning
4. Schedule and Release
5. Shop Floor Documents
6. Goods Issue
7. Completion Confirmation
8. Goods Receipt
9. Order Settlement

Financial Accounting (FI)
It is designed to collect transactional data for preparing the standard portfolio of reports. These reports are primarily, but not exclusively, directed at external parties.

Organizational Units:

- **Chart of Accounts** – a classification scheme consisting of a group of general ledger (G/L) accounts used by one or more company codes.
- **Business Area** – represents a separate area of operations or responsibilities within an organization and to which value changes recorded in Financial Accounting can be allocated.

Master Data:

- **General Ledger (G/L) Accounts** – the data storage area created by the unique combination of Company Code and Chart of Account. G/L lists the transactions affecting each account in the Chart of Accounts and the respective account balance.

Processes:

1. **Balance Sheet** – the summary of an organization's assets, liabilities, and equity at a point in time.
2. **Income Statement** – shows an organization's revenues and expenses at a point in time.
3. **Statement of Cashflows** - considered the associated changes in cash, which can be viewed as the most important of all assets.

Controlling (CO)

It is designed to collect transactional data for preparing internal reports to support decision-making within the enterprise. It provides methods for slicing and dicing data to view costs from an internal management perspective and gives a view of profitability beyond basic financial reporting.

Organizational Unit:

- **Operating Concern** – represents a part of an organization for which the sales market is structured uniformly.

Master Data:

- **Profit Center** – responsible for revenue generation and cost containment. It is evaluated on profit or return on investment.
- **Cost Center** – responsible for cost containment but not for revenue generation. One or more value-added activities are performed within each cost center.
- **Internal Order** – a temporary cost collector responsible for cost containment but not for revenue generation. It is used to plan, collect, and monitor the costs associated with a distinct short-term event, activity, or project.

Processes:

1. **Distribution** – a method for periodically allocating primary cost elements. The primary cost elements maintain their identities in sending and receiving objects.
2. **Assessment** – a method for allocating both primary and secondary cost elements. The primary and cost elements are grouped and transferred to receiver cost centers using a secondary cost element.

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